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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester B.Tech (CBCS) Degree Examinations

May 2024

CY1003-1 – MATERIALS CHEMISTRY FOR COMPUTER SYSTEMS

(AD, AM, CB, CC, CS, IS, RI)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. Potential of the calomel electrode depends on concentration of
A) Hg B) Hg_2Cl_2
C) KCl D) KOH
2. Which of the following is an example for ion selective electrode
A) Platinum electrode B) Gas electrode
C) Calomel electrode D) Glass electrode
3. Which of the following is suitable for powering portable devices such as smartphones and laptops?
A) Lead-acid battery B) Lithium-ion battery
C) Methanol-oxygen fuel cell D) Leclanché cell
4. What is the main component of the electrolyte in a sodium-ion battery
A) NaPF_6 B) NaCl
C) NaSO_4 D) Na_2CO_3
5. The total number of functional groups or bonding sites present in a monomer molecule is called.....
A) Functionality B) Degree of polymerization
C) Molecular weight D) Polymer
6. If degree of polymerization of polyethylene is 100, then the molecular weight is.....
A) 2000 B) 2500
C) 2800 D) 1500
7. Polymer composites are made of two components, namely.....
A) Fibre and matrix B) Matrix and silicone
C) Fibre and butyl rubber D) butyl rubber and silicone
8. sensor used for colorimetric estimation of copper.
A) Thermometric B) Gas
C) Optical D) Electro chemical
9. Anode used in the Electrochemical DO sensor.....
A) Au B) Ag
C) Pt D) Zn
10. In an electrochemical sensor electrode acts like a/an.....
A) Modifier B) Amplifier
C) Transducer D) Detector
11.type of coating is used in the preparation of printed circuit boards.
A) Electroplating B) Electroless plating
C) Inorganic coating D) Organic coating
12. Which among the following is a Flash memory device?
A) Hard Disc Drive (HDD) B) CD Drive
C) RAM D) USB drive

13. A ROM is
 A) Read Out Memory
 B) Read Once Memory
 C) Read Only Memory
 D) Read One Memory
14. Liquid crystal molecules have structure
 A) Rod like
 B) Square
 C) Cylindrical
 D) Triangular
15. Which of the following televisions deliver the best picture quality?
 A) LCD
 B) Plasma
 C) 3D
 D) OLED
16. Which of the following component make e-waste hazardous in nature?
 A) Lead
 B) Glass
 C) Plastic
 D) Iron
17. What does e- waste stand for?
 A) Environment waste
 B) Electronic waste
 C) Equipment waste
 D) None of the above
18. What is a good way to dispose of e-waste?
 A) Burn it in your backyard
 B) Give it to a reliable e-waste recycling company
 C) Throw it in the street
 D) Break it into small pieces and dispose
19. A Solar cell is an electrical device that converts the energy of light directly into electricity by effect:
 A) Physical
 B) Chemical
 C) Photovoltaic
 D) Photosynthesis
20. Materials used for solar cell is.....
 A) Silicon
 B) Copper
 C) Iron
 D) Sodium

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the construction and working of calomel electrode.	06	L*2	1	1
	b) Describe the construction and working of Li-ion battery.	06	L2	1	1
	c) Explain the determination of pH of an unknown solution using glass electrode.	04	L2	1	1
2.	a) Explain the synthesis, properties, and applications of carbon fibre.	07	L2	1	1
	b) A polymer has the following composition: 30 molecules of molecular mass 5000g/mol, 40 molecules of molecular mass 6000 g/mol and 50 molecule of molecular mass 8000g/mol. Calculate the number average and weight average molecular mass.	06	L3	1	2
	c) Define the following: i) specific conductance, ii) emf, iii) conducting polymer.	03	L1	1	1
3.	a) Define concentration cell? The emf of the cell $\text{Cd}/\text{CdSO}_4(0.093\text{M})//\text{CdSO}_4(x\text{ M})/\text{cd}$ is 0.086V at 25°C. Find the value of x.	04	L3	1	2
	b) Explain the mechanism in conduction of polyaniline.	06	L2	1	1
	c) Explain the construction and working of methanol-oxygen fuel cell.	06	L2	1	1
Unit – II					
4.	a) Explain the working principle of conductometric and optical sensors.	06	L2	2	1
	b) Explain the application of electrochemical sensor for sensing SO _x gas.	04	L2	2	1
	c) Describe the electroless plating of copper on PCB.	06	L2	2	1
5.	a) What are photoactive and electroactive materials. Explain their working principle in display system.	06	L2	2	
	b) Describe the electro-optic effect of liquid crystals.	06	L2	2	1
	c) Give any two properties and application of OLED.	04	L1	2	1

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| 6. a) | Describe the principle and working of electrochemical sensor and mention its applications. | 06 | L2 | 2 | 1 |
| b) | Mention any three properties and applications of QLED. | 06 | L1 | 2 | 1 |
| c) | Mention the classification of electronic memory devices | 04 | L1 | 2 | 1 |

Unit – III

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|-------|--|----|----|---|---|
| 7. a) | Mention the sources of e-waste and describe the need for e-waste management. | 06 | L2 | 3 | 1 |
| b) | Describe the extraction of gold from e-waste. | 06 | L2 | 3 | 1 |
| c) | What is green fuel? Mention any three advantages of green fuel. | 04 | L1 | 3 | 1 |
| 8. a) | Describe the construction and working of solar photovoltaic cell. | 06 | L2 | 3 | 1 |
| b) | Give any four advantages of recycling. | 04 | L1 | 3 | 1 |
| c) | Discuss the extraction of e-waste by pyro metallurgical process. | 06 | L2 | 3 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
